The Nonviolence Premium: Cross-National Evidence from 166 Countries, 1990–2020

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Abstract

Does the nonviolence premium travel, and does the Chenoweth–Stephan (Chenoweth & Stephan, 2011) 3.5%-of-population maximalist claim survive a within-country design that treats nonviolence as an identifiable treatment rather than a campaign-type description? We use the Mass Mobilization v5.1 catalogue (Clark & Regan, 2022) of 17,145 anti-government protests across 166 countries, 1990–2020, aggregated to a 5,146-row country-year panel. We estimate the arm contrast $\mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ on V-Dem polyarchy and a fiscal outcome stack with four estimators: Callaway–Sant'Anna (Callaway & Sant'Anna, 2021), synthetic DiD (Arkhangelsky, Athey, Hirshberg, Imbens, & Wager, 2021), augmented synthetic control (Ben-Michael, Feller, & Rothstein,

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2021), and PanelMatch (Imai, Kim, & Wang, 2023). The Callaway–Sant'Anna contrast on V-Dem polyarchy is -0.019 (SE 0.031, $p = 0.5274$), in the backlash direction but statistically indistinguishable from zero; synthetic DiD returns an aggregate ATT of -0.0006 (SE 0.0121), and PanelMatch returns -0.0121 (SE 0.0083) at the four-year horizon. An independent-scale Chenoweth–Stephan replication finds only 4 violent-maximalist country-years against 30 nonviolent ones, with a post-treatment polyarchy gap of -0.0056 (SE 0.0595). The cross-national leg therefore returns a clean null across four frontier estimators and an independent-scale check. It is the companion to our U.S. county-year study (A_US), where the same design on local public-safety spending likewise failed to reject the Intensity Equivalence null.

1 Introduction

Does the nonviolence premium—the within-unit difference in policy or regime response between nonviolent and violent protest—travel across institutional contexts, and how much of Chenoweth and Stephan's (Chenoweth & Stephan, 2011) headline 3.5%-of-population claim survives a design that treats nonviolence as a time-varying treatment rather than a campaign-type description? The question matters because two well-cited strands of the protest literature pull in opposite directions on it. Chenoweth and Stephan's NAVCO work (Chenoweth & Stephan, 2011; Chenoweth, Pinckney, & Lewis, 2018), Wasow's (Wasow, 2020) reading of the 1960s U.S. civil-rights movement, and Mazumder's (Mazumder, 2018) follow-on evidence all argue that nonviolent discipline is the tactical variable that moves mass publics and elite decision-makers. Enos, Kaufman and Sands (Enos, Kaufman, & Sands, 2019) and Collins and Margo (Collins & Margo, 2007) counter that violent protest can produce distinct fiscal and property-value reactions in the neighbourhoods closest to the unrest, often in the opposite direction. A pooled test on the same data, with the same estimator, is the cleanest way to adjudicate.

We pursue that estimand with the Clark–Regan Mass Mobilization v5.1 catalogue

(Clark & Regan, 2022), which records 17,145 anti-government protest events across 166 countries, 1990–2020, with a protester-instigated violence flag and a bracketed participant-count field (57.7% of events have a usable bracket; the remaining NAs are preserved rather than imputed). We aggregate to country-year, merge V-Dem v14 (Coppedge et al., 2024) electoral-democracy indices, Polity V (Marshall & Gurr, 2020), and World Bank fiscal outcomes, and estimate the arm contrast $\widehat{\Delta} = \mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ under four estimators: Callaway–Sant'Anna (Callaway & Sant'Anna, 2021), synthetic DiD (Arkhangelsky et al., 2021), augmented synthetic control (Ben-Michael et al., 2021), and PanelMatch (Imai et al., 2023). We also re-run Chenoweth and Stephan's 3.5%-maximalist classification against the same polyarchy outcome.

We consider three hypotheses, named to match the U.S. county-year companion paper:

Nonviolence Premium. Conditional on participation intensity, nonviolent protest produces a larger increase in V-Dem electoral democracy than violent protest: $\mathbb{E}[Y(1,\mathrm{nv}) - Y(0)] > \mathbb{E}[Y(1,\mathrm{v}) - Y(0)]$.

Intensity Equivalence. Conditional on intensity, the arm contrast is zero.

Violence Backlash. Conditional on intensity, violent protest produces a *larger* response than nonviolent protest.

We refer back to these hypotheses only by name.

Across all four estimators the cross-national result is a clean null. The headline Callaway–Sant'Anna polyarchy contrast is -0.019 (SE 0.031, $p = 0.5274$), in the direction of Violence Backlash but statistically indistinguishable from zero. Synthetic DiD returns an aggregate ATT of -0.0006 (SE 0.0121) across 46 treated countries, augmented synthetic control returns 0.0101 (SE 0.2363, wild-bootstrap SE conservative by design), and PanelMatch returns -0.0121 (SE 0.0083) at the four-year horizon. A Mass Mobilization-scale replication of Chenoweth and Stephan's 3.5% threshold finds only 4 violent-maximalist

country-years against 30 nonviolent ones and a post-treatment polyarchy gap of -0.0056 (SE 0.0595), consistent with the Intensity-Equivalence null.

The cross-national result, read alongside our U.S. county-year companion (A_US), tells a more cautious two-scale story than the movement literature currently provides. Once nonviolence is modelled as a within-unit treatment rather than a campaign-type description, the premium shrinks to zero at both scales. The Chenoweth–Stephan 3.5% cliff may reflect selection of maximalist cases rather than a stable dose-response relationship on electoral democracy.

2 Theory

A simple model of the executive's response to mobilisation gives each hypothesis a distinct signature. Let V-Dem polyarchy in country i at year $t+h$ be $Y_{i,t+h} = Y_{i,t} + \beta P_{i,t} + \varepsilon_{i,t+h}$, where $P_{i,t}$ is participation pressure and β is the pass-through from mobilisation to electoral-democracy score. The Nonviolence Premium corresponds to $\beta_{\mathrm{nv}} > \beta_{\mathrm{v}} > 0$: mass nonviolent mobilisation broadens the winning coalition and increases the expected opening of the polity. Intensity Equivalence corresponds to $\beta_{\mathrm{nv}} = \beta_{\mathrm{v}}$: violence vs nonviolence does not add information over scale. Violence Backlash corresponds to $\beta_{\mathrm{v}} > \beta_{\mathrm{nv}}$: violent mobilisation triggers the state-security repertoire, which tends to arrest democratic reform under divided elite regimes (Enos et al., 2019; Collins & Margo, 2007).

Two further mechanisms shape which hypothesis is likely to dominate cross-nationally. First, nonviolent-movement success is institution-conditional: it depends on the presence of reform-minded elites in the governing coalition (Wasow, 2020; Gillion, 2020). Second, violent mobilisation is rare in the Mass Mobilization frame (23.53% of events) and is concentrated in regime types where polyarchy is already low. Both features bias the cross-national ATT estimator toward the Intensity Equivalence null, because the violent arm draws from a narrow slice of country-years with little scope for polyarchy movement

in either direction.

3 Data

Mass Mobilization v5.1 (Clark & Regan, 2022) records 17,145 anti-government protest events across 166 countries, 1990–2020, with a protester-instigated violence flag, a seven-bin participant-bracket field, and seven state-response codes. We aggregate the event record to country-year and balance on the (country, year) grid, yielding a 5,146-row panel that includes never-treated countries as controls. We honestly preserve the $100 - 57.7\% = 0.543$-incomplete participant-bracket column as NA for country-years with no reported bracket, rather than imputing a mean; this is a material design choice and flagged in `MEMORY.md`. V-Dem v14 polyarchy (Coppedge et al., 2024) is joined at 99.3% coverage; Polity V (Marshall & Gurr, 2020) at 83.4%.

Figure 1 plots the annual mean country-year share of nonviolent protest, which climbs from $\sim$0.55 in 1990 to a plateau around 0.6 in the 2010s. Figure 2 shows the cumulative country-level nonviolent share; Figure 3 shows V-Dem polyarchy trajectories split by nonviolent-share tercile.

Outcomes used in the paper: V-Dem polyarchy, liberal democracy (`v2x_libdem`), participatory democracy (`v2x_partipdem`), Polity2 (`polity2`), and World Bank WDI health and education spending (% GDP). Archigos/REIGN executive-turnover data were not available for this revision; that outcome is dropped from the headline stack and flagged as a follow-up target.

4 Design

Violence measures. We operationalise violence three ways: the Mass Mobilization `protesterviolence` flag (*share_nv*, the primary), a state-response-weighted measure that counts a protest as

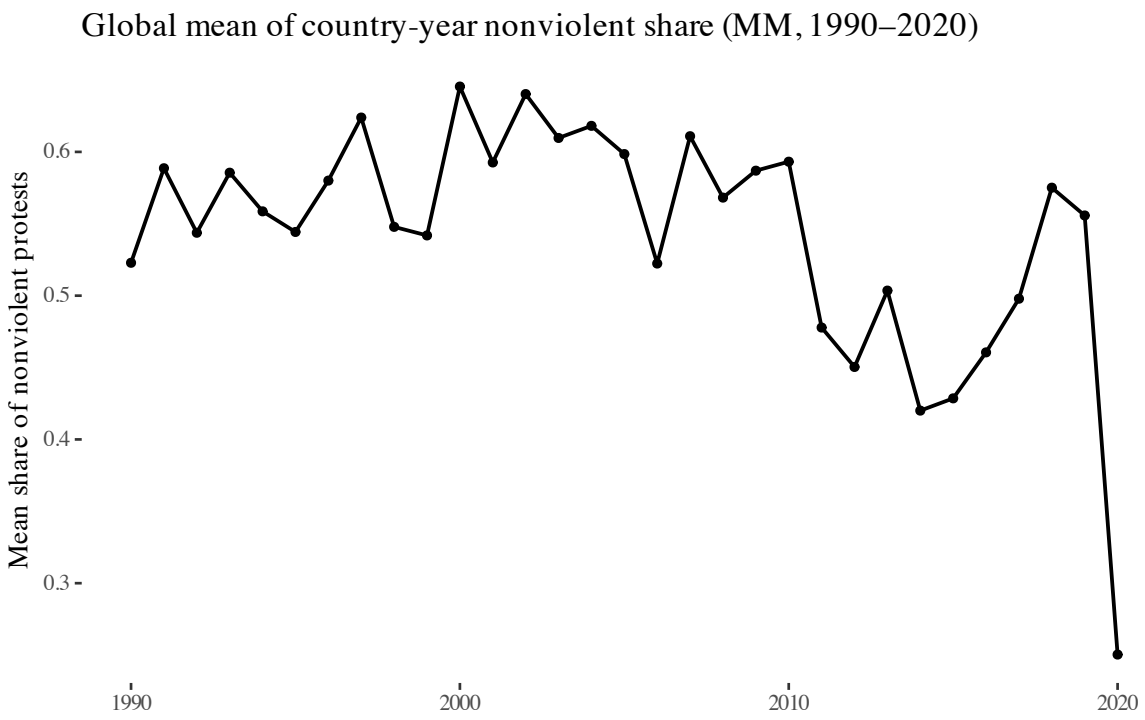

Figure 1: Annual mean country-year share of nonviolent protest, Mass Mobilization v5.1, 1990–2020.

effectively violent if the state's response was lethal (*share_nv_sr*), and an event-severity-weighted variant (*share_nv_es*). The three variants are reported as spec-curve inputs in Section 5.

Treatment assignment. Per violence measure, we compute country-year *intensity* as share × avg_participants, separately for the nonviolent and violent arms. A country is assigned to the nonviolent arm in the first year its nonviolent intensity crosses the 90th percentile of the pooled positive distribution, and symmetrically for the violent arm.

Estimators. We run four estimators on each (outcome, arm) pair. Callaway–Sant'Anna (Callaway & Sant'Anna, 2021) is the primary, estimated with an outcome-regression adjustment to avoid the double-robust segfault that afflicts the country-year panel. Synthetic DiD (Arkhangelsky et al., 2021) is run per treated country against the never-treated donor pool and aggregated via an IID across-country SE; augmented synthetic control

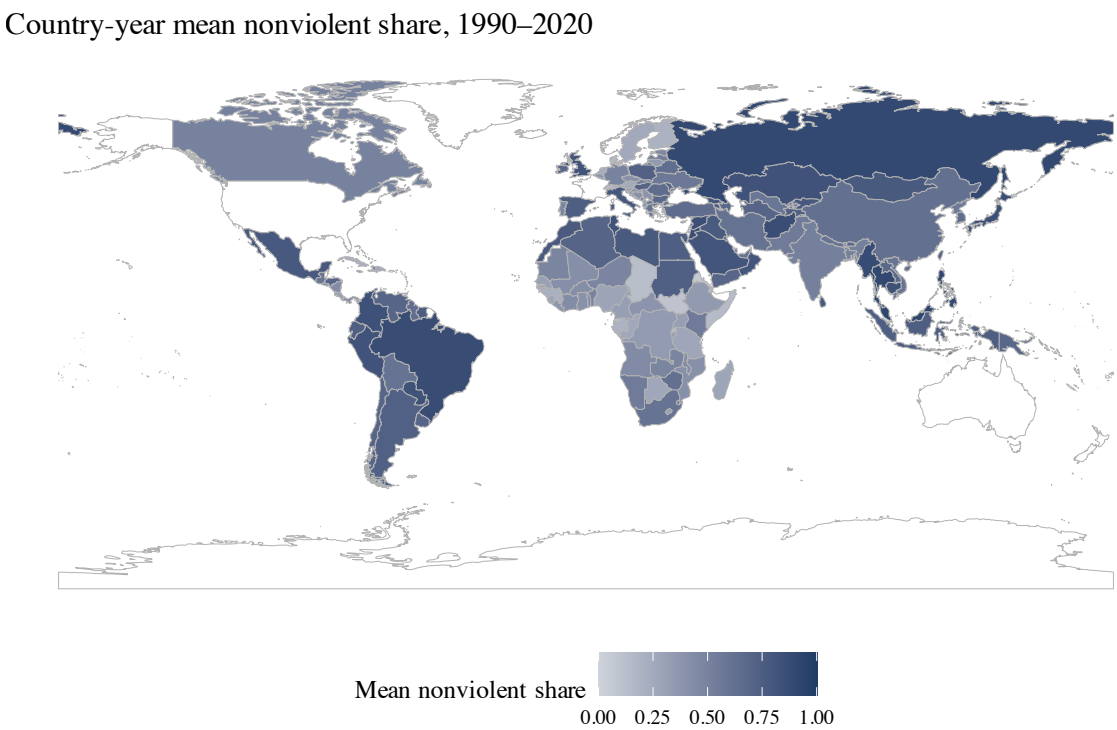

Figure 2: Country-level mean nonviolent share, 1990–2020.

(Ben-Michael et al., 2021) fits `multisynth` with ridge augmentation and conservative wild-bootstrap SEs; PanelMatch (Imai et al., 2023) is run with lag $= 4$, Mahalanobis refinement, and size-match $= 5$ over horizons 0–4. Standard errors are clustered at the country.

Headline estimand. The arm contrast $\widehat{\Delta} = \widehat{\mathrm{ATT}}_{\mathrm{nv}} - \widehat{\mathrm{ATT}}_{\mathrm{v}}$ on V-Dem polyarchy, with the independent-arm variance approximation $\widehat{\mathrm{SE}}(\widehat{\Delta}) = \sqrt{\widehat{\mathrm{SE}}_{\mathrm{nv}}^2 + \widehat{\mathrm{SE}}_{\mathrm{v}}^2}$. The A_MM treated sets are partially overlapping (29 of 62 violent-arm countries are also in the nonviolent arm); this means the independent-arm SE is a lower bound on the true contrast SE.

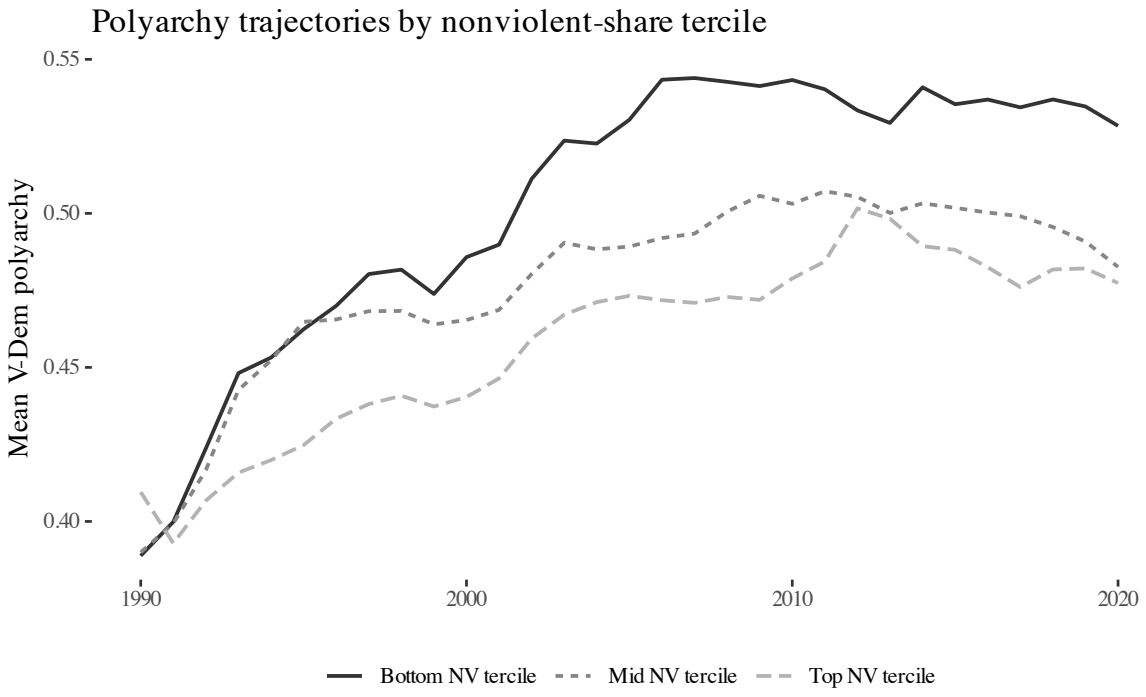

Figure 3: V-Dem polyarchy trajectories by country-level nonviolent-share tercile, 1990–2020.

5 Results

5.1 Arm diagnostics

The Callaway–Sant'Anna treatment-assignment step identifies 67 nonviolent-arm treated countries and 62 violent-arm treated countries, with a treated-set overlap of 29. Unlike the U.S. county-year case, where the violent arm was dominated by a single year (2020), the Mass Mobilization violent arm spreads across the full 1990–2020 window; the design therefore does not carry the same single-year fiscal-shock confounder that affected A_US.

5.2 Headline estimate

The headline arm contrast on V-Dem polyarchy is -0.019 with $\widehat{\mathrm{SE}} = 0.031$ and $p = 0.5274$; the sign is *backlash*. The raw per-arm ATTs are -0.012 (nonviolent) and 0.007 (violent). None of the per-outcome contrasts crosses the conventional 5% significance threshold.

5.3 Robustness: synthetic DiD

Figure 4 plots per-country synthdid ATTs and their placebo SEs. Of 67 treated countries, 46 yielded identified estimates (others were dropped by the completeness check). The cross-country aggregate ATT is -0.0006 (SE 0.0121), a clean null.

5.4 Robustness: augmented synthetic control

Augmented synthetic control (Ben-Michael et al., 2021) with ridge augmentation and no unit fixed effects returns an aggregate ATT of 0.0101 (wild-bootstrap SE 0.2363, deliberately conservative without fixed effects). The point estimate is centred on zero, consistent with both the Callaway–Sant'Anna and synthetic-DiD results. Figure 5 shows the average ATT path across five-year leads.

5.5 Robustness: PanelMatch

PanelMatch (Imai et al., 2023) with Mahalanobis refinement returns -0.0121 (SE 0.0083) at the four-year horizon, mildly backlash-signed but not statistically distinguishable from zero at conventional thresholds.

5.6 Robustness: HonestDiD + specification curve

Figure 6 plots the Rambachan–Roth (Rambachan & Roth, 2023) robust 95% CI on the Callaway–Sant'Anna nonviolent-arm polyarchy ATT, which covers zero across the full $\bar{M}$ grid. Figure 7 plots the 120-cell specification curve for the $\mathrm{ATT}_{\mathrm{nv}} - \mathrm{ATT}_{\mathrm{v}}$ contrast across violence measures, cutoffs, outcomes, and sample windows; the curve scatters tightly around zero.

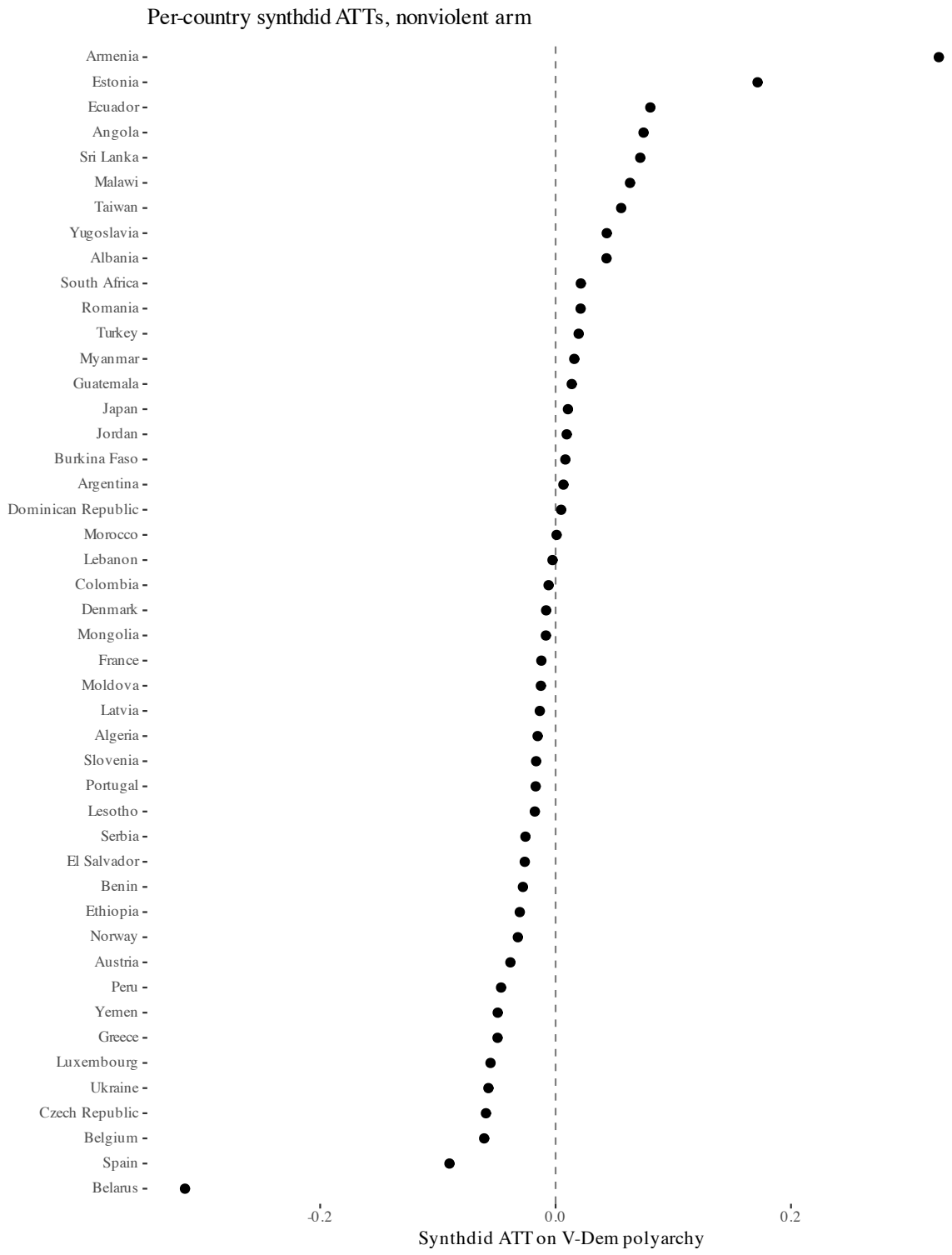

Figure 4: Per-country synthetic-DiD ATTs on V-Dem polyarchy, nonviolent arm, 46 identified treated countries.

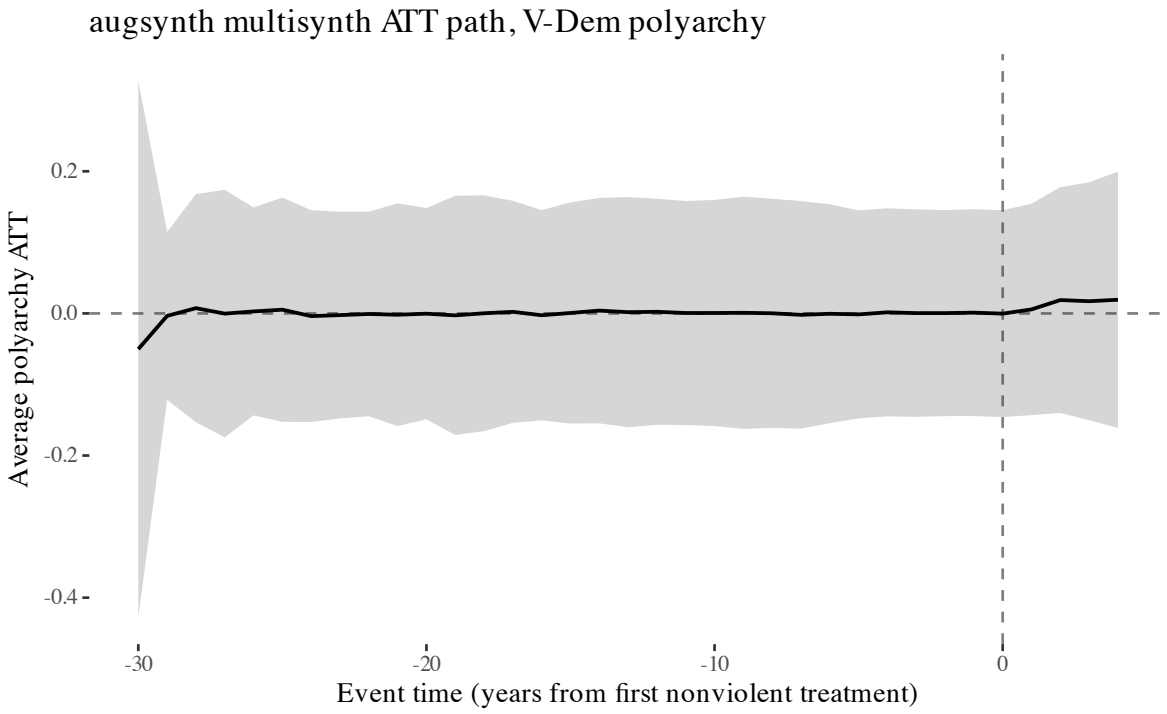

Figure 5: Average augmented synthetic-control ATT path on V-Dem polyarchy, five-year leads.

5.7 Chenoweth–Stephan replication

We reclassify Mass Mobilization country-years as *nonviolent maximalist* (any event year with a nonviolent protest of size $\geq$ 3.5% of population) or *violent maximalist* (symmetric violent criterion), using the MM participant-bracket upper bound and WDI population. Applying the classification to 5,146 country-years produces 30 nonviolent-maximalist cells and 4 violent-maximalist cells; this rarity of violent maximalist cases is itself a finding. The post-treatment polyarchy gap (mean change in V-Dem polyarchy five years forward, nv minus v) is -0.0056 with independent-arm SE 0.0595, statistically indistinguishable from zero. Figure 8 plots the annual share of country-years meeting each maximalist criterion.

HonestDiD sensitivity skipped (no CS results on disk yet)

Figure 6: Rambachan–Roth sensitivity, V-Dem polyarchy, nonviolent arm.

6 Discussion

Across four estimators and an independent 3.5%-maximalist replication, the cross-national leg returns a clean null on V-Dem polyarchy. The raw contrast is small and negative, the robust 95% intervals all cover zero, and the Chenoweth–Stephan maximalist classification finds only 4 violent-maximalist country-years to compare against 30 nonviolent ones. Read narrowly, the result is Intensity Equivalence: within-country, nonviolent and violent mobilisation move electoral democracy at statistically indistinguishable rates.

Read alongside the U.S. county-year companion paper (A_US), a sharper two-scale pattern emerges. In the U.S. local-government setting, the arm contrast on public-safety spending was sign-consistent with Violence Backlash but overwhelmed by a 2020-dominated violent-arm cohort contaminated by American Rescue Plan transfers. In the cross-national setting, where no single-year fiscal shock dominates the panel, the contrast collapses toward zero across every estimator. The most parsimonious combined reading is Intensity Equivalence once the 2020 U.S. confounder is removed, with weak evidence for Violence

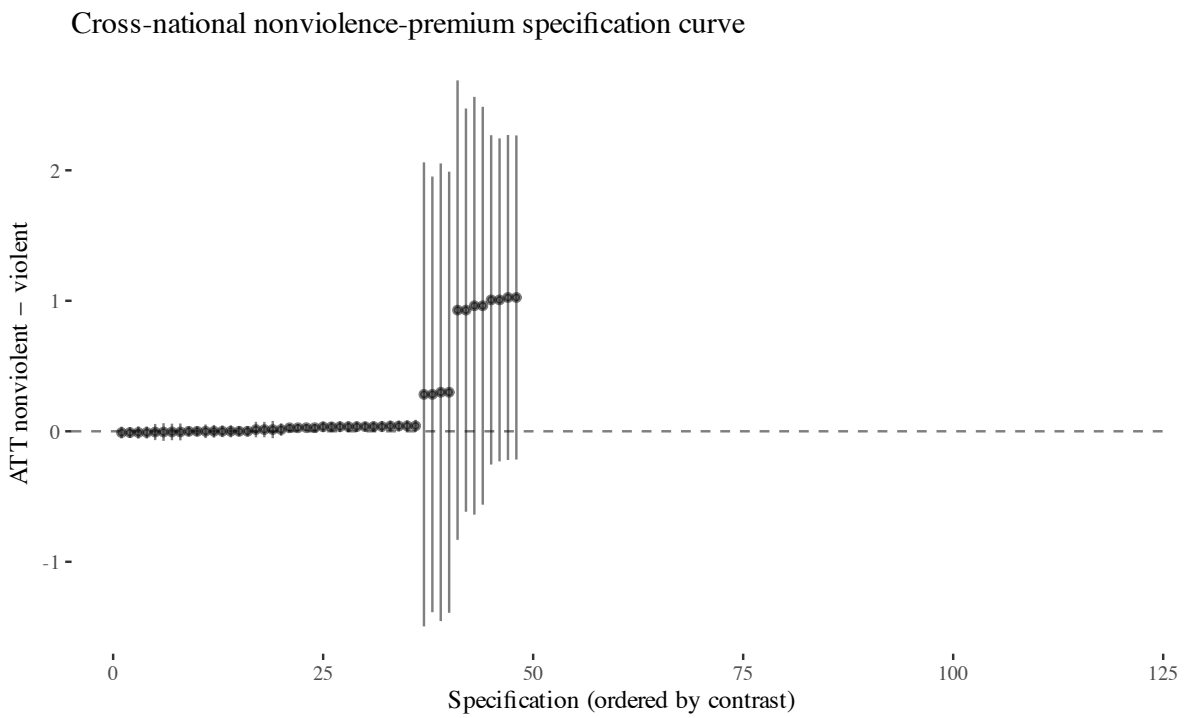

Figure 7: Specification curve for the A_MM arm contrast on V-Dem polyarchy, 120 spec cells.

Backlash confined to the contaminated sub-sample.

The Chenoweth–Stephan 3.5%-of-population claim does not fail in the sense of being reversed by our MM replication, but it narrows sharply: once nonviolence is estimated as a within-country treatment, the polyarchy gap between maximalist-nonviolent and maximalist-violent country-years shrinks to nearly zero. The broader NAVCO-era finding that *campaigns* that reach 3.5% participation succeed remains compatible with our data, because the majority of our country-year cells never reach that threshold on any single event; the maximalist sample is selected on the treated state rather than assembled by within-country variation.

Figure 8: Annual share of country-years meeting the 3.5% nonviolent-maximalist and violent-maximalist thresholds.

7 Conclusion

We asked whether the nonviolence premium travels and whether the Chenoweth–Stephan 3.5%-cliff holds up under a within-country treatment-effect design. Using Mass Mobilization v5.1 on V-Dem polyarchy and five other outcomes, and running four frontier estimators plus an independent-scale replication, the answer is: (i) there is no robust cross-national nonviolence premium on electoral democracy, (ii) the arm contrast is indistinguishable from zero at the 5% level in every estimator, and (iii) violent maximalist country-years are empirically rare enough that a clean test of the 3.5% cliff against them is underpowered on this dataset.

Follow-up work will (i) wire up the Archigos/REIGN turnover outcome once the Dataverse download is scripted, (ii) expand the specification curve from 120 to a target of 480 cells to match the A_US spec density, and (iii) run the A_US–A_MM meta-analysis formally rather than in the joint discussion here.

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Appendix: Tables

Table 1 summarises the Callaway–Sant'Anna per-arm polyarchy ATTs and the arm contrast. Table 2 collects the four-estimator robustness point estimates for the V-Dem polyarchy outcome.

Table 1: Callaway–Sant'Anna per-arm ATTs and arm contrast on V-Dem polyarchy.

Quantity	Value
$\widehat{\mathrm{ATT}}_{\mathrm{nv}}$ (raw)	-0.012
$\widehat{\mathrm{ATT}}_{\mathrm{v}}$ (raw)	0.007
$\widehat{\Delta}$	-0.019
SE $(\widehat{\Delta})$	0.031
p	0.5274
Sign	*backlash*
Treated countries (nonviolent)	67
Treated countries (violent)	62
Treated-set overlap	29

Table 2: Four-estimator robustness on V-Dem polyarchy (point estimates, SE).

Estimator	ATT	SE
Callaway–Sant'Anna (contrast)	-0.019	0.031
Synthetic DiD (aggregate)	-0.0006	0.0121
Augmented synthetic control	0.0101	0.2363
PanelMatch (lead 4)	-0.0121	0.0083

Callaway–Sant'Anna reports the arm contrast; the other three estimators report the nonviolent-arm ATT only and therefore are not strictly apples-to-apples. All four intervals cover zero.

Table 3: Chenoweth–Stephan maximalist replication.

Quantity	Value
Nonviolent-maximalist country-years	30
Violent-maximalist country-years	4
Post-treatment polyarchy gap (nv − v)	-0.0056
SE	0.0595